Makhura Kamogelo

Week 3 Practicals

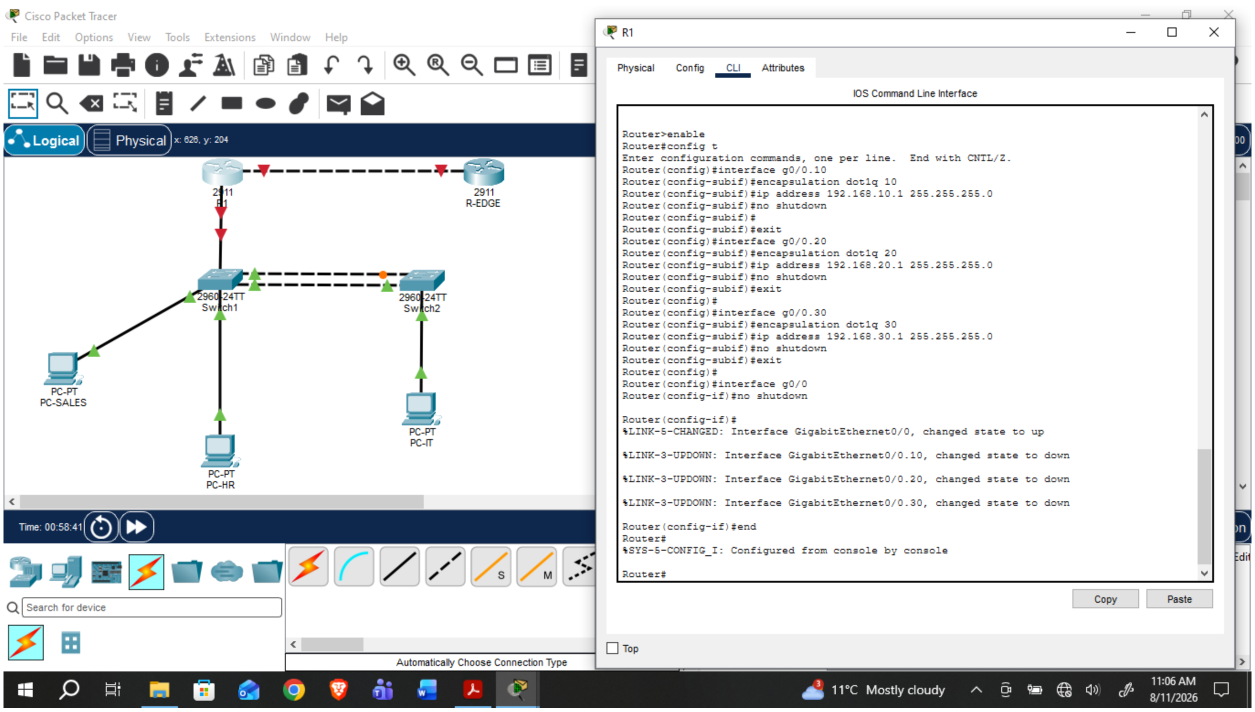
Tasks:

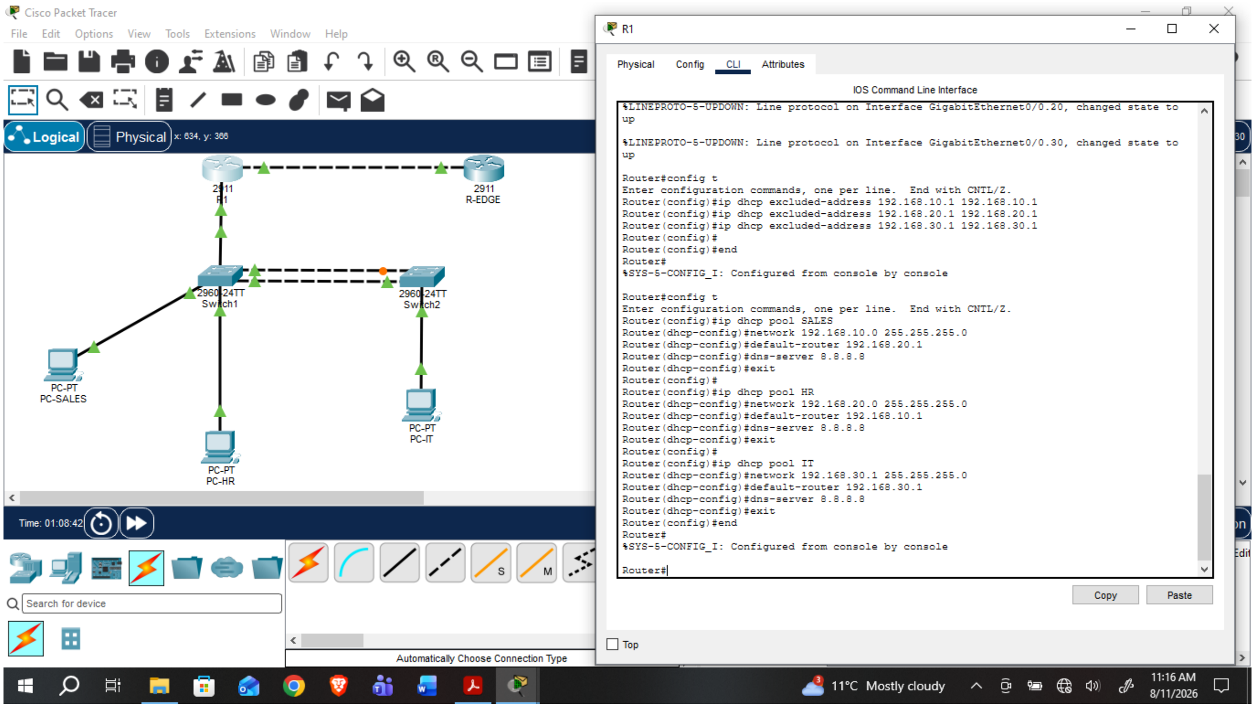
Step 1: Configure DHCP Pools on R1

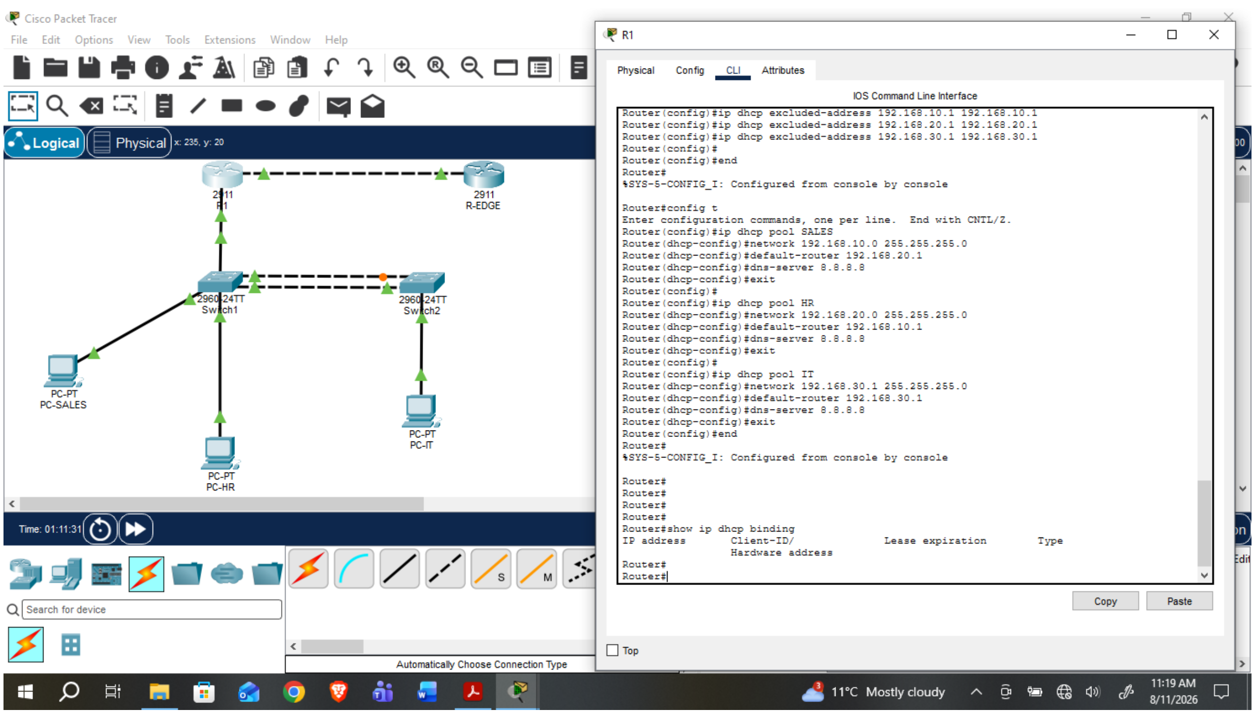
1. ip dhcp excluded-address 192.168.10.1 192.168.10.1 ip dhcp excluded-address 192.168.20.1 192.168.20.1 ip dhcp excluded-address 192.168.30.1 192.168.30.1

2. ip dhcp pool SALES network 192.168.10.0 255.255.255.0 default-router 192.168.10.1 dns-server 8.8.8.8

3. Repeat for the HR (192.168.20.0) and IT (192.168.30.0) pools.



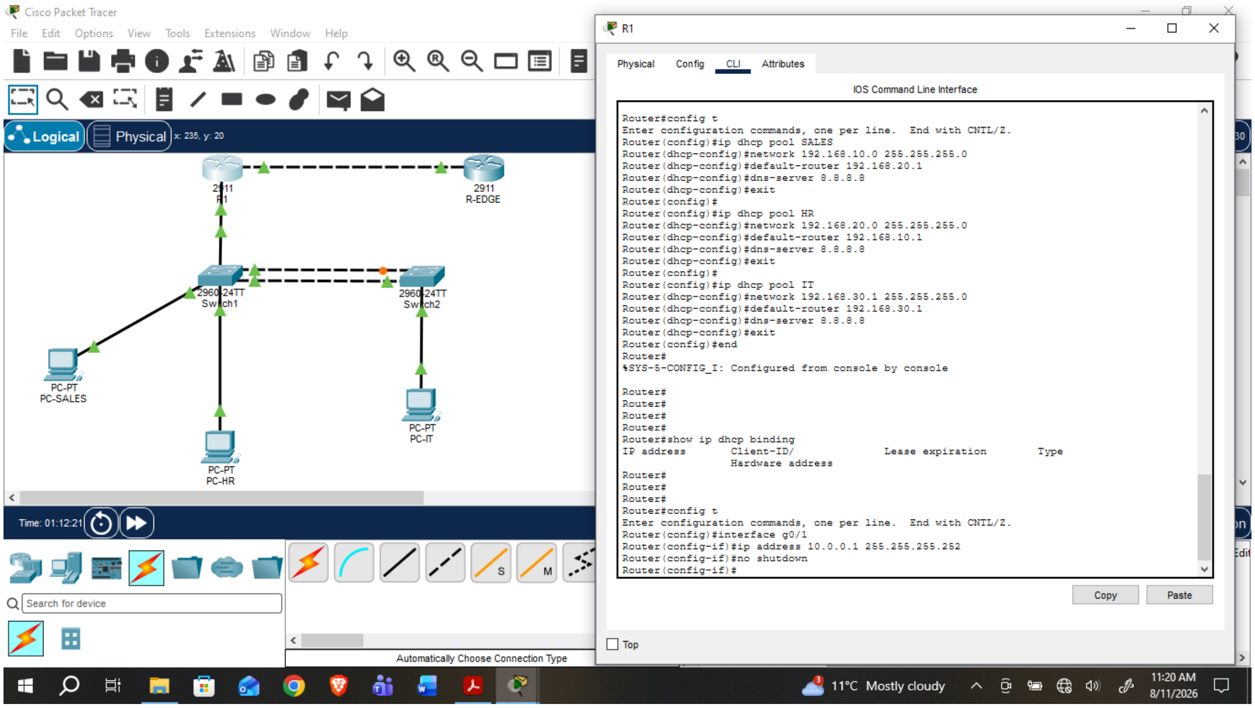

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 634, y: 366
2911
R1
2911
R-EDGE
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Time: 01:08:42
Search for device
Automatically Choose Connection Type
R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.30, changed state to up
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.1
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Copy
Paste
Top
11°C Mostly cloudy
11:16 AM
8/11/2026


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 235, y: 20
2911
R1
2911
R-EDGE
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Time: 01:11:31
Search for device
Automatically Choose Connection Type
R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(config)#ip dhcp excluded-address 192.168.10.1 192.168.10.1
Router(config)#ip dhcp excluded-address 192.168.20.1 192.168.20.1
Router(config)#ip dhcp excluded-address 192.168.30.1 192.168.30.1
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool SALES
Router(dhcp-config)#network 192.168.10.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.20.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool HR
Router(dhcp-config)#network 192.168.20.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#
Router#show ip dhcp binding
IP address   Client-ID/   Lease expiration   Type
                 Hardware address
Router#
Router#
Copy
Paste
Top
11°C Mostly cloudy
11:19 AM
8/11/2026

Step 2: Test DHCP

1. Set each PC's IP configuration to DHCP and verify it receives an address automatically.

2. show ip dhcp binding

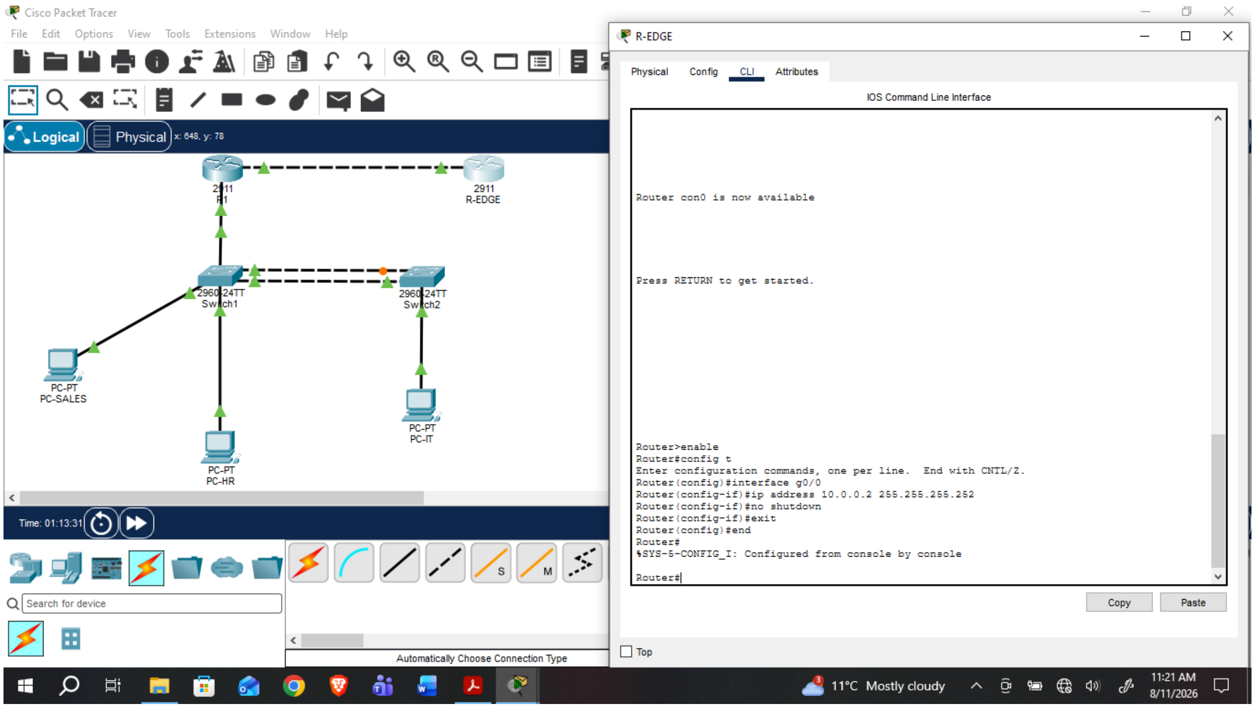


Step 3: Static & Default Routing to R-EDGE

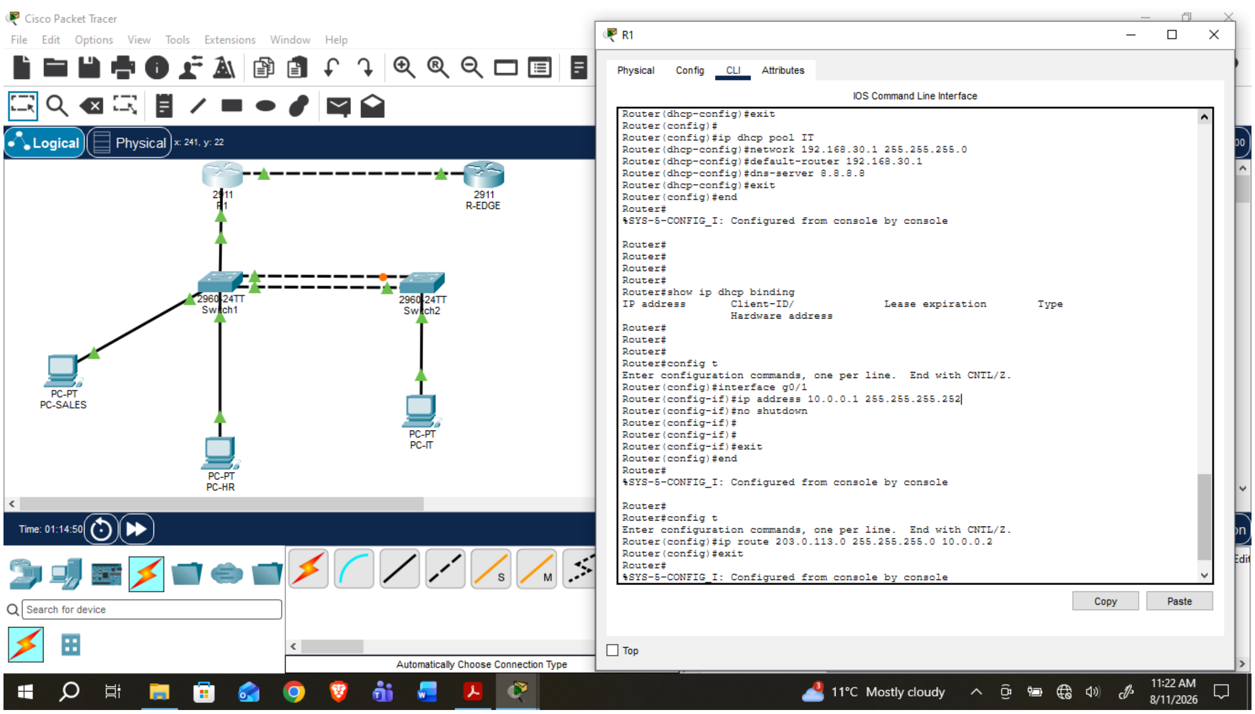
1. Assign IP addresses on the R1–R-EDGE link.

2. On R1: ip route [R-EDGE's LAN network] [mask] [next-hop]

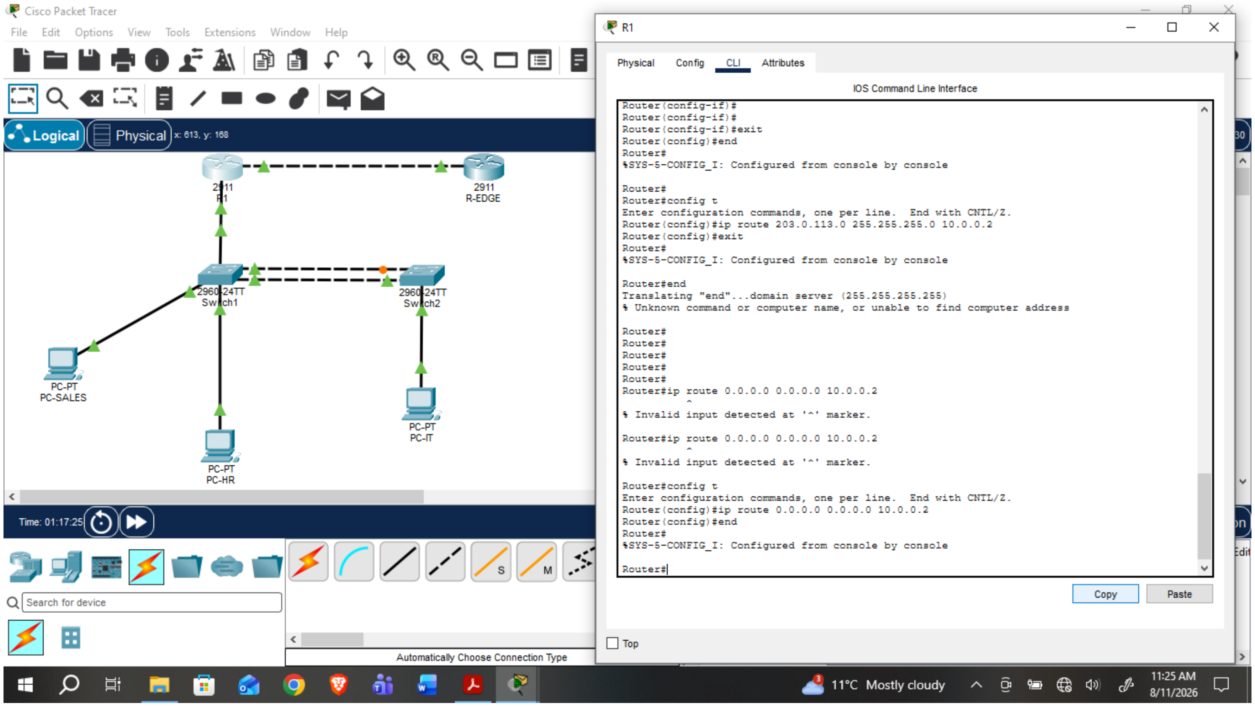
3. On R1: ip route 0.0.0.0 0.0.0.0 [R-EDGE's interface IP] (default route)



Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
2911
R1
2911
R-EDGE
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Time: 01:13:31
Search for device
Automatically Choose Connection Type
R-EDGE
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router con0 is now available
Press RETURN to get started.
Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Copy
Paste
Top
11°C Mostly cloudy
11:21 AM
8/11/2026

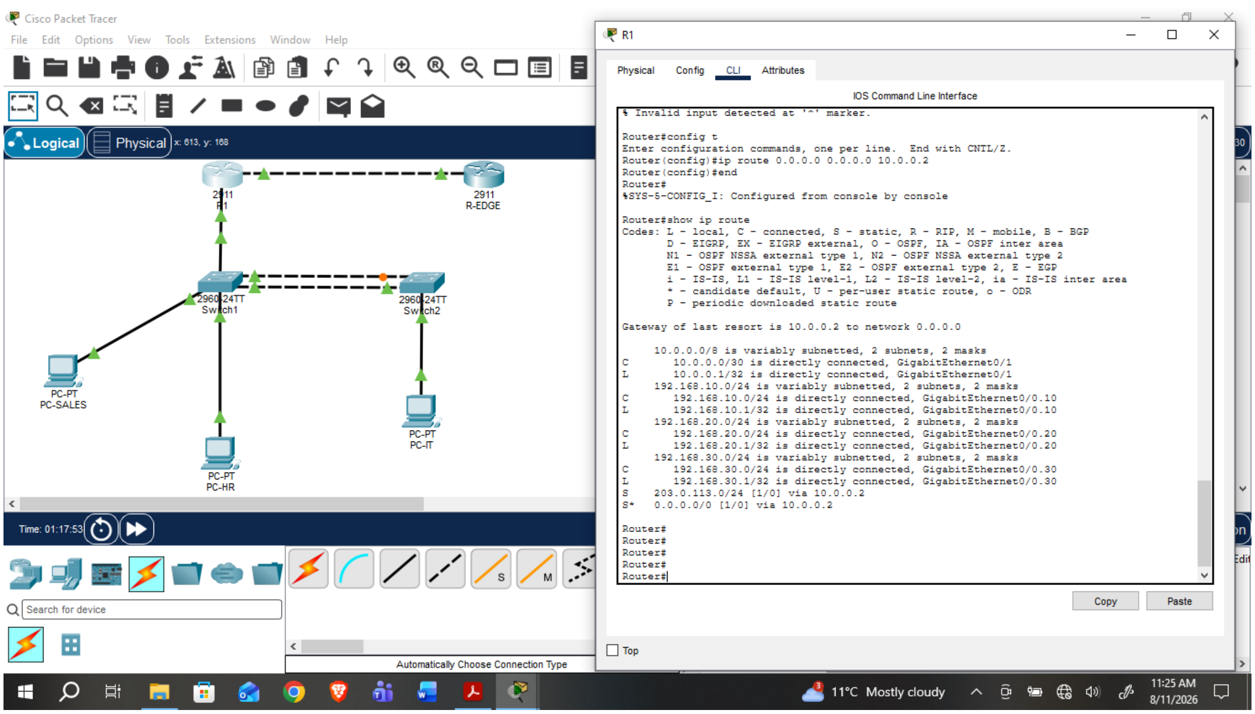


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
2911
R1
2911
R-EDGE
2960-24TT
Switch1
2960-24TT
Switch2
PC-PT
PC-SALES
PC-PT
PC-HR
PC-PT
PC-IT
Time: 01:14:50
Search for device
Automatically Choose Connection Type
R1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Router(dhcp-config)#exit
Router(config)#
Router(config)#ip dhcp pool IT
Router(dhcp-config)#network 192.168.30.1 255.255.255.0
Router(dhcp-config)#default-router 192.168.30.1
Router(dhcp-config)#dns-server 8.8.8.8
Router(dhcp-config)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#
Router#
Router#
Router#show ip dhcp binding
IP address   Client-ID/   Lease expiration   Type
                 Hardware address
Router#
Router#
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface g0/1
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#no shutdown
Router(config-if)#
Router(config-if)#
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 203.0.113.0 255.255.255.0 10.0.0.2
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
Copy
Paste
Top
11°C Mostly cloudy
11:22 AM
8/11/2026



Step 4: Verify

1. show ip route



Reflection Questions (answer these in your submission):

1. What's the risk of not excluding the gateway address from a DHCP pool?

If you don’t exclude 192.168.10.1, 192.168.20.1, 192.168.30.1, DHCP could lease those gateway IPs to a PC. That causes an IP address conflict.

The router and the PC would both claim the same gateway address. Devices in that VLAN would lose their default gateway, so inter-VLAN routing and Internet access would break for everyone in that subnet until the conflict is resolved.

You may see intermittent connectivity and difficult-to-troubleshoot network problems, Devices may lose network or Internet connectivity, and the router/gateway and the client may have the **same IP address**.